ECON 0150 | Homework 4.1

Due: Friday, Nov. 13 at 5PM

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Answer clearly and completely. You are welcomed and encouraged to work in groups so long as your work is your own. Submit your figures and answers to Gradescope.

Q1. Unemployment and BMI

The article *Zhang, Q., Lamichhane, R., & Wang, Y. (2014)* in the Journal of Clinical Medicine studies the associations between US adult obesity and mental health and state/county economic conditions in recessions. The Homework 4.1 Notebook loads a sample of individual-level data from the 2011 Behavioral Risk Factor Surveillance System (BRFSS), which includes the following variables:

- `BMI5` – individual Body Mass Index (kg/m^2)
- `MENTHLTH` – days of poor mental health in the past 30 days
- `unemployment_rate` – county-level unemployment rate (in percentage points)

Using this data, estimate the following linear model:

$$\texttt{BMI5}_i = \beta_0 + \beta_1 \cdot \texttt{unemployment_rate}_i + \varepsilon_i \tag{1}$$

a) Report the estimated coefficients $\hat{\beta}_0$ and $\hat{\beta}_1$, and the p-value on $\hat{\beta}_1$.

b) State the null hypothesis for β_1.

c) Interpret your estimated coefficient $\hat{\beta}_1$ in words. What does the estimate suggest about the relationship between county unemployment and individual BMI?

Q2. Unemployment and Mental Health

The 2011 BRFSS also includes information on mental health. The variable `MENTHLTH` records the number of days in the past 30 days when the respondent's mental health was "not good." Estimate the following linear model:

$$\texttt{MENTHLTH}_i = \beta_0 + \beta_1 \cdot \texttt{unemployment_rate}_i + \varepsilon_i \tag{2}$$

a) Report and interpret the estimated coefficient $\hat{\beta}_1$.

b) Write a complete interpretation of the p-value for $\hat{\beta}_1$ from your model in part (a). What exactly does it tell us in words?

Q3. Reading a Regression Output

A researcher wants to test whether temperature (in °F) predicts daily ice cream sales (in dollars) using data from 60 days at a local shop. Their regression output shows:

```
                 coef  std err        t    P>|t|   [0.025   0.975]
------------------------------------------------------------------
Intercept      -159.47   42.300   -3.771    0.000  -244.17   -74.77
temperature       4.78    0.540    8.852    0.000     3.70     5.86
------------------------------------------------------------------
```

a) Write the regression model being estimated.

b) Interpret the intercept coefficient (-159.47) in context. Does it have a meaningful interpretation here?

c) Interpret the slope coefficient (4.78) in context.

d) What is the null hypothesis being tested for the `temperature` coefficient?

e) Interpret the p-value for the `temperature` coefficient in words.

f) If the temperature on a given day is 80°F, what does the model predict for ice cream sales?